Prabu Shankar P
ROLL NO 241501142

EXPERIMENT 07

Integration with Zoho CRM using APIs

Aim

Sync owner contacts to Zoho CRM with an authenticated connection and retry-safe handling.

Procedure

1. Use a test CRM organization approved by your institution. Send only owner name and email, not service history.
2. Create a Creator connection library_crm (or vehicle_crm) with the scope ZohoCRM.modules.contacts.WRITE. Never paste raw tokens into scripts.
3. Use Email as the duplicate-check key. Map Owner_Name to Last_Name, Email to Email, returned ID to CRM_Contact_ID.
4. Mark a record Pending after save; call the upsert after save or from a scheduled queue so servicing never depends on CRM being up.
5. Check both HTTP errors and the record-level status in the response before setting Synced; on error store a sanitized message.
6. Re-run the same request and confirm only one contact exists.

```
contact = Map(); contact.put("Last_Name", rec.Owner_Name); contact.put("Email", rec.Email);
payload = Map(); payload.put("data", {contact}); payload.put("duplicate_check_fields", {"Email"});
resp = invokeurl [ url :"https://www.zohoapis.com/crm/v8/Contacts/upsert"
  type :POST parameters:payload.toString() connection:"vehicle_crm" ];
```

Use the API domain for your own data center. Add response and exception handling before enabling retries.

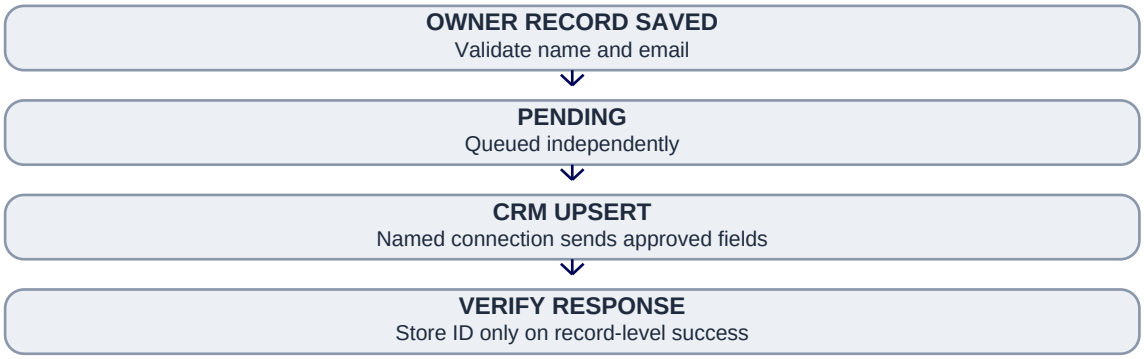
Implementation and verification

Test	Expected evidence
Valid owner sync	One contact, ID stored, status Synced
Repeat request	No duplicate
Invalid email	Readable error, not Synced
Revoked connection	Authorization failure logged, no secrets
CRM outage	Retry or Pending; service flow unaffected

Evidence and testing

E07: field mapping, redacted connection config, sanitized response, CRM verification, retry log.

Process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

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Result

Expected result and verification criteria. A test owner syncs once, repeats do not duplicate, and failures are visible and recoverable.